

Submodularity and Submodular Fns

Def'n: E be a finite ground set

$f: 2^E \rightarrow \mathbb{R}$ is called submodular

if: $\forall A, B \subseteq E: \underline{f(A) + f(B) \geq f(A \cup B) + f(A \cap B)}$

Def'n: Marginal Value

For $A \subseteq E: f_A(i) = f(A + i) - f(A).$

Thm: T.F.A.E.

1. f is submodular
2. $\forall A \subseteq B \subseteq E, \forall i \in E \setminus B: f_A(i) \geq f_B(i)$
3. $\forall A \subseteq E, \forall i, j \in E \setminus A$

$$f_A(i) \geq f_{A+j}(i)$$

Definitions and Properties

Defn: A submodule fn $f: 2^E \rightarrow \mathbb{R}$ is

called monotone: if $A \subseteq B$,
 $f(A) \leq f(B)$.

Lemma: $\alpha, \beta \geq 0$, f, g submodular $\Rightarrow \alpha f + \beta g$ is submodular.

Lemma: If f is submodular, g linear
 $\Rightarrow f - g$ is submodular.

Lemma: If f, g submodular, if $f - g$ monotone
 $\Rightarrow \min(f, g)$ submodular.

Some Examples

i. Rank fns of matroids

ii. Capacity of cuts: $D = (V, A)$, $c: A \rightarrow \mathbb{R}_+$

Fix $r, s \in V$

For $S \subseteq V \setminus \{r, s\}$, $f(S) \triangleq c(\delta^+(S \cup \{r\}))$

Exercise: f is submodular.

iii. Number of neighbors

$G = (V, E)$, $V = V_1 \cup V_2$, $\forall u \in V$

$\Gamma(u) = \text{neighbors of } u$. Exercise: $f(u) = |\Gamma(u)|$ is submodular.

iv. Mixed rank of two matroids

$M_1 = (E, \mathcal{X}_1)$, $M_2 = (E, \mathcal{X}_2)$ $f(S) = r_1(S) + r_2(E \setminus S)$ is submodular.

Recall: Matrix Int. Thm — $\min_{S \subseteq E} f(S) = \max_{\mathcal{I} \in \mathcal{X}_1, \mathcal{X}_2} |\mathcal{I}|$



More Examples